Department of Computing

CS 419: Deep Learning (2+1)

Class: BSCS-13

Project: Plant Disease Classification Task Using Deep Learning & Transfer Learning

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Learning Outcome

It's a group project. Please mention the contribution of each member clearly in the notebook.

Contents

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1) Objectives

With successful completion of this project, you will be able to:

- Independently design a deep learning pipeline using transfer learning.
- Explore, fine-tune, and compare different CNN architectures.
- Develop an intuition for model performance on real-world image classification tasks.
- Reflect on design choices, model selection, and tuning strategies.
- Communicate your results through clean code and thoughtful analysis.

2) Core Ideas

- Transfer Learning refers to using pre-trained models to address new but related problems with limited labeled data.
- The difference between simple feature extraction and fine tuning is that in feature extraction we use pre-trained CNNs as static feature generators; however, in the case of fine tuning we unfreeze some layers to adapt pre-trained weights to the new task.
- Choosing two diverse CNN families encourages you to think about architecture differences (depth, width, receptive field, computational cost).

3) Project Tasks & Deliverables

Plant diseases threaten food security globally. Early and accurate disease detection using AI methodologies can significantly reduce crop loss. Your task is to build a system that classifies plant diseases using deep learning and transfer learning techniques.

3a) Dataset

Use the following dataset available on Kaggle:

New Plant Diseases Dataset

Upload the dataset to your Google Drive or work entirely in Google Colab/kaggle. Be creative in managing data size and class balance.

3b) Tasks

You will build upon your previous lab work and extend it into a more comprehensive comparative study

Choose 3 CNN Architectures

Select a total of three CNN architectures for this project.

You have already implemented two architectures in the **Open-Ended Lab (Lab 10)**, each from a different family (e.g., ResNet vs Inception, MobileNet vs DenseNet, EfficientNet vs VGG). For this project, you are required to extend your work by selecting a third architecture.

The third architecture must be from a **different design family** than the previously selected models (not just a variant of an existing one).

Briefly justify your choice of all three architectures, highlighting their key differences and explaining why each was selected for this task.

Apply Transfer Learning in Two Ways

Apply transfer learning (feature extraction and fine-tuning) to all three architectures. Reuse results from the open-ended lab for the first two models and extend the same setup to the third
i) As a feature extractor (freeze convolutional base).
ii) With fine-tuning (unfreeze and train upper layers).

Perform Comparative Analysis

Compare models in terms of accuracy, generalization, training stability, and computational efficiency.

Use identical data splits, preprocessing steps, augmentation, training epochs, and evaluation metrics across all models unless explicitly justified.

Provide reasoning for observed differences based on architectural characteristics and training behavior.

Visual Interpretation (Optional)

Use tools like Grad-CAM or Saliency Maps to visualize what your models "see". This will help you in understanding model decision boundaries and failure cases.

3c) Deliverables

The submission includes:

- A well-structured notebook including dataset overview, methodology, experiments, results, and comparative analysis
- A final report under (Lab 13 submission)